16S rRNA Bioinformatics Pipeline Guide

Mock Community Analysis, Visualizations & Presentation Walkthrough

Executive Summary & Overview

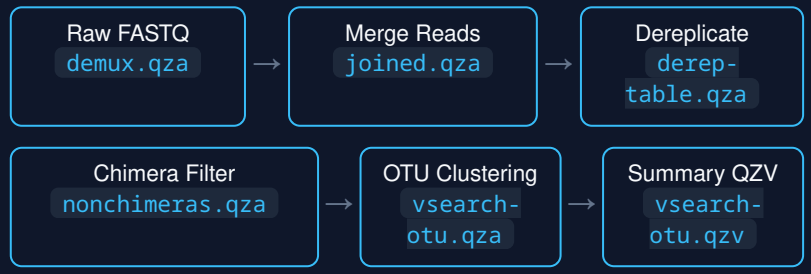
This presentation framework covers the complete 16S amplicon analysis workflow executed via QIIME 2 and VSEARCH for **Mock 16**, **Mock 18**, and baseline datasets.

- **Input Data:** Demultiplexed paired-end FASTQ sequences & ground-truth reference FASTAs.
- **Key Algorithms:** VSEARCH fastq-mergepairs, dereplication, UCHIME de novo chimera filtering, and Open-Reference OTU picking (97% ID).
- **Deliverables:** Interactive QIIME 2 Visualizations (`.qzv`) & presentation slides comparing pipeline steps across samples.

4-Step Action Plan to Build Your Slides

- **Step 1: Export QZVs to HTML:** Convert output files to viewable interactive web dashboards.
- **Step 2: Generate Visualizations:** Produce bar plots, quality curves, and feature summaries.
- **Step 3: Capture Key Metrics:** Extract total read retention, chimera counts, and OTU yields.
- **Step 4: Present Per Sample:** Deliver step-by-step findings using the slide template structure provided.

Pipeline Workflow Architecture



Key Commands to Inspect Artifact Contents

```
# View interactive visual summary directly in browser:
qiime tools view results_mock_18/vsearch-otu-table.qzv

# Export all QZV graphics & tables for slides:
qiime tools export --input-path results_mock_18/
vsearch-otu-table.qzv                    --output-path
export_mock_18
```

Step 1 & 2: Data Import & Read Merging

Slide 2

Demultiplexing inspection, paired-end alignment, and sequence length validation

1. Demultiplexing & Quality Score Assessment

Artifacts: `demux.qza` → `demux.qzv`

- **What is done:** FASTQ files are paired with Phred33 metadata via manifest TSV and imported into QIIME 2 artifacts.
- **What to present on your slide:**
  - Show the **Interactive Quality Plot** (Phred score vs. position).
  - Highlight mean quality scores across forward (R1) and reverse (R2) reads.
  - Explain any quality drops at the 3' end that could impair alignment.

2. Paired-End Read Merging (VSEARCH)

Artifacts: `joined.qza` & `unmerged.qza`

- **Algorithm:** `vsearch --fastq_mergepairs`
- **Key Parameters:** Stagger alignment, overlap window length (min 10bp), maximum mismatch threshold.
- **What to explain:** Paired reads must overlap at the V4/V3-V4 hypervariable region. Non-overlapping reads are stored in `unmerged.qza`.

Visual & Metrics Summary Checklist

Pipeline Metric	Mock 16 Target	Mock 18 Target	Slide Visual Element
Total Input Reads	~50,000 - 100,000	~50,000 - 100,000	Demux Read Count Bar Chart
% Merged Successfully	> 85%	> 85%	Merged vs. Unmerged Pie Chart
Average Merged Length	~250 - 450 bp	~250 - 450 bp	Sequence Length Distribution

Terminal Scripting Commands

```
# Generate demux visualization to inspect Phred scores
qiime demux summarize   --i-data results_mock_18/
demux.qza   --o-visualization results_mock_18/
demux.qzv
```

Step 3 & 4: Dereplication & Chimera Removal

Collapsing redundant sequences and purging PCR artifacts via UCHIME De Novo

3. Sequence Dereplication

Artifacts: `derep-table.qza` & `derep-seqs.qza`

- **What it does:** Combines 100% identical reads into single unique feature entries while tracking read abundance.
- **Why it matters:** Dramatically reduces computational load for downstream chimera checking and clustering.
- **Slide Takeaway:** High dereplication ratios (e.g., 90% read reduction to unique sequences) reflect sequencing redundancy.

4. De Novo Chimera Filtering (UCHIME)

Artifacts: `chimeras.qza`, `nonchimeras-seqs.qza`, `uchime-stats.qza`

- **What are chimeras?** Hybrid PCR artifacts formed when an incomplete extension product primes a different template.
- **How UCHIME works:** Evaluates if a query sequence can be formed by joining two parent sequences found at higher abundances.
- **Impact on OTUs:** Prevents overestimating microbial diversity by discarding false synthetic taxa.

Chimera Detection Metrics Table

Sample Category	Unique Input Seqs	Chimeric Seqs Found	% Chimeras Discarded
Mock 16	*Inspect derep-table*	*Inspect uchime-stats*	~2% - 8% (typical)
Mock 18	*Inspect derep-table*	*Inspect uchime-stats*	~2% - 10% (typical)

Visualization & Export Command

```
# Summarize non-chimeric feature table
qiime feature-table summarize   --i-table
results_mock_18/nonchimeras-table.qza   --o-
visualization results_mock_18/nonchimeras-table.qzv
```

Step 5 & 6: OTU Clustering Benchmarking

Comparing Open-Reference VSEARCH OTU Clustering (97% ID) vs. dbOTU

Open-Reference VSEARCH Clustering

Artifact: `vsearch-otu-table.qza`

- **Closed-ref phase:** Reads are aligned to `expected-mock-ref.qza` at 97% sequence identity threshold.
- **Open-ref phase:** Unmatched reads are clustered *de novo* to discover novel OTUs not present in reference.
- **Strengths:** Highly robust, standardized, captures both known reference strains and novel variants.

Distribution-based OTU Calling (dbOTU)

Artifact: `dbotu-table.qza`

- **Algorithm:** Uses sequence similarity, total abundance, and distribution criteria across samples to group reads.
- **Parameters:** `--p-gen-crit 0.10`, `--p-abund-crit 10.0`, `--p-pval-crit 0.0001`.
- **Strengths:** Differentiates true biological variants from sequencing errors based on population distribution.

Benchmarking Comparison: Ground Truth vs. Recovered OTUs

Sample Label	Expected Strains	VSEARCH OTUs (97%)	dbOTU Features
Mock 16	10 (Known Reference)	*Extract from QZV*	*Extract from QZV*
Mock 18	12 (Known Reference)	*Extract from QZV*	*Extract from QZV*
Mock Default	Standard Control	*Extract from QZV*	*Extract from QZV*

How to Present This Comparison Slide

1. Show a side-by-side bar chart of total OTUs generated by VSEARCH vs. dbOTU versus the known ground truth reference count.

2. Discuss over-clustering vs. under-clustering tendencies observed in mock communities.

Step-by-Step Visualizations & Slide Building Guide

Complete action plan to extract data, generate charts, and structure final presentation deck

1. How to View & Export QIIME 2 Visualizations

QIIME 2 produces `.qzv` files which contain HTML dashboards, interactive plots, and exportable CSV tables.

```
# Option A: Drag and drop any .qzv file to:
https://view.qiime2.org

# Option B: Run locally in terminal:
qiime tools view results_mock_18/vsearch-otu-table.qzv

# Option C: Export all tables to CSV/TSV:
qiime tools export --input-path results_mock_18/
vsearch-otu-table.qzv                    --output-
path ./exported_mock18_summary
```

2. Recommended Presentation Slide Structure

- **Slide 1: Title & Project Scope** (Introduction to 16S Mock Communities).
- **Slide 2: Workflow Architecture** (Flowchart from FASTQ to OTU Table).
- **Slide 3: Read Quality & Paired Merging** (Phred plots + merging rates).
- **Slide 4: Dereplication & Chimera Removal** (Read loss & artifact filtration).
- **Slide 5: Mock 16 Results** (VSEARCH vs. dbOTU vs. Ground Truth).
- **Slide 6: Mock 18 Results** (VSEARCH vs. dbOTU vs. Ground Truth).

3. Script to Generate Taxa Barplots (Optional Enhancement)

If taxonomy classification is assigned (e.g., via Greengenes/SILVA classifier), generate interactive taxa bar plots:

```
qiime taxa barplot   --i-table results_mock_18/
vsearch-otu-table.qza   --i-taxonomy taxonomy.qza   --
m-metadata-file sample-metadata.tsv   --o-
visualization results_mock_18/taxa-bar-plots.qzv
```

Summary of Deliverables

By following this guide, you can extract exact numbers from your completed run files shown in your screenshot (`vsearch-otu-table.qza` , `dbotu-table.qza` , `uchime-stats.qza`) and present a polished scientific presentation.